Week 12: Forecasting Volatility

Consider the GARCH(1,1) model

$$\begin{aligned} r_t &= \mu + \varepsilon_t, \\ \varepsilon_t &= \sigma_t u_t, \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \end{aligned}$$

Assume

$$E(u_t \mid \mathcal{F}_{t-1}) = 0 \qquad \text{and} \qquad E(u_t^2 \mid \mathcal{F}_{t-1}) = 1.$$

Let $\mathcal{F}_t$ denote the information set available at time t. We want to derive the h-step-ahead forecast

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t).$$

Assume that

$$\alpha_0 > 0, \qquad \alpha_1 \geq 0, \qquad \beta_1 \geq 0.$$

For stationarity we require

$$\alpha_1 + \beta_1 < 1.$$

For $h \geq 2$ show that

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

Using the Law of Iterated Expectations,

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E\left[E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) \mid \mathcal{F}_t\right].$$

Since

$$\varepsilon_{t+h-1} = \sigma_{t+h-1} u_{t+h-1},$$

we have

$$\varepsilon_{t+h-1}^2 = \sigma_{t+h-1}^2 u_{t+h-1}^2.$$

Therefore, the inner expectation becomes

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) = E(\sigma_{t+h-1}^2 u_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}).$$

Given $\mathcal{F}_{t+h-2}$, the conditional variance σ^2_{t+h-1} is known because

$$\sigma^2_{t+h-1} = \alpha_0 + \alpha_1 \varepsilon^2_{t+h-2} + \beta_1 \sigma^2_{t+h-2}.$$

Hence,

$$E(\sigma^2_{t+h-1} u^2_{t+h-1} \mid \mathcal{F}_{t+h-2}) = \sigma^2_{t+h-1} E(u^2_{t+h-1} \mid \mathcal{F}_{t+h-2}).$$

Since

$$E(u^2_{t+h-1} \mid \mathcal{F}_{t+h-2}) = 1,$$

it follows that

$$E(\varepsilon^2_{t+h-1} \mid \mathcal{F}_{t+h-2}) = \sigma^2_{t+h-1}.$$

Substituting this into the LIE expression,

$$\begin{aligned} E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) &= E\left[E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) \mid \mathcal{F}_t\right] \\ &= E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Thus,

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

One-Step-Ahead Forecast

For time $t + 1$, the conditional variance is

$$\sigma_{t+1}^2 = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+1}^2 \mid \mathcal{F}_t) &= E\left(\alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2 \mid \mathcal{F}_t\right) \\ &= \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2. \end{aligned}$$

Therefore,

$$E(\sigma_{t+1}^2 \mid \mathcal{F}_t) = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2.$$

This is because ε_t^2 and σ_t^2 are known at time t.

Two-Step-Ahead Forecast

For time $t + 2$,

$$\sigma_{t+2}^2 = \alpha_0 + \alpha_1 \varepsilon_{t+1}^2 + \beta_1 \sigma_{t+1}^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+2}^2 \mid \mathcal{F}_t) &= E\left(\alpha_0 + \alpha_1 \varepsilon_{t+1}^2 + \beta_1 \sigma_{t+1}^2 \mid \mathcal{F}_t\right) \\ &= \alpha_0 + \alpha_1 E(\varepsilon_{t+1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Two-Step-Ahead Forecast

Since

$$E(\varepsilon_{t+1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Substituting this into the two-step-ahead forecast,

$$\begin{aligned} E(\sigma_{t+2}^2 \mid \mathcal{F}_t) &= \alpha_0 + \alpha_1 E(\sigma_{t+1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Thus,

$$E(\sigma_{t+2}^2 \mid \mathcal{F}_t) = \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Three-Step-Ahead Forecast

For time $t + 3$,

$$\sigma_{t+3}^2 = \alpha_0 + \alpha_1 \varepsilon_{t+2}^2 + \beta_1 \sigma_{t+2}^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= E\left(\alpha_0 + \alpha_1 \varepsilon_{t+2}^2 + \beta_1 \sigma_{t+2}^2 \mid \mathcal{F}_t\right) \\ &= \alpha_0 + \alpha_1 E(\varepsilon_{t+2}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+2}^2 \mid \mathcal{F}_t). \end{aligned}$$

Three-Step-Ahead Forecast

Since

$$E(\varepsilon_{t+2}^2 \mid \mathcal{F}_t) = E(\sigma_{t+2}^2 \mid \mathcal{F}_t),$$

we obtain

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= \alpha_0 + \alpha_1 E(\sigma_{t+2}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+2}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+2}^2 \mid \mathcal{F}_t). \end{aligned}$$

Therefore,

$$E(\sigma_{t+3}^2 \mid \mathcal{F}_t) = \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+2}^2 \mid \mathcal{F}_t).$$

Three-Step-Ahead Forecast

Using the two-step-ahead forecast,

$$E(\sigma_{t+2}^2 \mid \mathcal{F}_t) = \alpha_0 + (\alpha_1 + \beta_1)E(\sigma_{t+1}^2 \mid \mathcal{F}_t),$$

we get

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= \alpha_0 + (\alpha_1 + \beta_1)\left[\alpha_0 + (\alpha_1 + \beta_1)E(\sigma_{t+1}^2 \mid \mathcal{F}_t)\right] \\ &= \alpha_0 + \alpha_0(\alpha_1 + \beta_1) + (\alpha_1 + \beta_1)^2 E(\sigma_{t+1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0\left[1 + (\alpha_1 + \beta_1)\right] + (\alpha_1 + \beta_1)^2 E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

General h-Step-Ahead Forecast

For $h \geq 2$, the GARCH equation is

$$\sigma_{t+h}^2 = \alpha_0 + \alpha_1 \varepsilon_{t+h-1}^2 + \beta_1 \sigma_{t+h-1}^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+h}^2 \mid \mathcal{F}_t) &= E\left(\alpha_0 + \alpha_1 \varepsilon_{t+h-1}^2 + \beta_1 \sigma_{t+h-1}^2 \mid \mathcal{F}_t\right) \\ &= \alpha_0 + \alpha_1 E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t). \end{aligned}$$

General h-Step-Ahead Forecast

Since

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

Substituting this result into the forecast equation gives

$$\begin{aligned} E(\sigma_{t+h}^2 \mid \mathcal{F}_t) &= \alpha_0 + \alpha_1 E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Therefore, for $h \geq 2$,

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

Closed-Form Solution

Let

$$\rho = \alpha_1 + \beta_1.$$

Then, for $h \geq 2$, the recursive forecast is

$$E(\sigma^2_{t+h} \mid \mathcal{F}_t) = \alpha_0 + \rho E(\sigma^2_{t+h-1} \mid \mathcal{F}_t).$$

Starting from the one-step-ahead forecast,

$$E(\sigma^2_{t+1} \mid \mathcal{F}_t) = \alpha_0 + \alpha_1 \varepsilon^2_t + \beta_1 \sigma^2_t,$$

we obtain

$$E(\sigma^2_{t+2} \mid \mathcal{F}_t) = \alpha_0 + \rho E(\sigma^2_{t+1} \mid \mathcal{F}_t),$$

and

$$E(\sigma^2_{t+3} \mid \mathcal{F}_t) = \alpha_0 + \rho E(\sigma^2_{t+2} \mid \mathcal{F}_t).$$

Substituting recursively,

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= \alpha_0 + \rho \left[\alpha_0 + \rho E(\sigma_{t+1}^2 \mid \mathcal{F}_t)\right] \\ &= \alpha_0(1+\rho) + \rho^2 E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Continuing this pattern,

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \alpha_0 \left(1 + \rho + \rho^2 + \cdots + \rho^{h-2}\right) + \rho^{h-1} E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Geometric Series

Using the geometric series,

$$1 + \rho + \rho^2 + \cdots + \rho^{h-2} = \frac{1 - \rho^{h-1}}{1 - \rho},$$

General h-Step-Ahead Forecast

We now have

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \alpha_0 \frac{1-\rho^{h-1}}{1-\rho} + \rho^{h-1} E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Since $\rho = \alpha_1 + \beta_1$, this can be written as

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \alpha_0 \frac{1-(\alpha_1+\beta_1)^{h-1}}{1-(\alpha_1+\beta_1)} + (\alpha_1+\beta_1)^{h-1} E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Recall that the unconditional variance is

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}.$$

Since $\rho = \alpha_1 + \beta_1$, we have

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \rho}.$$

Therefore, the h-step-ahead forecast can be written as

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \bar{\sigma}^2 + (\alpha_1 + \beta_1)^{h-1} \left[E(\sigma_{t+1}^2 \mid \mathcal{F}_t) - \bar{\sigma}^2 \right].$$

As $h \to \infty$,

$$(\alpha_1 + \beta_1)^{h-1} \to 0,$$

provided that

$$\alpha_1 + \beta_1 < 1.$$

Hence,

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) \to \bar{\sigma}^2 \qquad \text{as } h \to \infty.$$

Therefore, GARCH variance forecasts mean-revert to the unconditional variance.

Stylised Facts

The ARCH(1) and GARCH(1,1) models capture several **stylised facts** of asset returns:

- **Volatility clustering:** large shocks tend to be followed by large shocks, and small shocks tend to be followed by small shocks.

- **Time-varying volatility:** the conditional variance changes over time depending on past shocks and/or past volatility.

Stylised Fact Not Captured by ARCH(1) and GARCH(1,1)

The standard ARCH(1) and GARCH(1,1) models do not capture the leverage effect, because positive and negative shocks of the same size have the same effect on future volatility.

For leverage effects, we usually need models such as GJR-GARCH or EGARCH.

Revision

MDS

Definition

A time series ε_t is an martingale difference sequence (MDS) if

$$E(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0$$

Here $\mathcal{F}_{t-1}$ denotes the information set available up to time $t-1$.

White Noise

A time series $\{\epsilon_t\}_{t\geq 1}$ is called a **white noise** process if

1. $\mathbb{E}[\epsilon_t] = 0$;
2. $\mathbb{V}[\epsilon_t] = \sigma_\epsilon^2$ for all t;
3. $\gamma_\epsilon(k) = \mathrm{Corr}(\epsilon_t, \epsilon_{t-k}) = 0$, for all t and k.

This is sometimes written as $\epsilon_t \sim \mathrm{WN}(0, \sigma_\epsilon^2)$.

independently and identically distributed (i.i.d.)

u_t is assumed to be independently and identically distributed for all t (i.i.d.), that is

$$u_t \sim \text{i.i.d. } (0, \sigma^2).$$

This notation means that u_t has mean zero and variance σ^2.

The assumption that u_t is independent over time means that it has no autocorrelation at any lag.

It also has a constant variance σ^2 over time so it is homoskedastic.

A time series that has no autocorrelation and is homoskedastic is sometimes called *white noise*. Therefore u_t is an example of a white noise process.

Stationary

A time series Y_t is defined to be covariance *stationary* if

1. $E(Y_t) = \mu$ does not depend on t
2. $\text{var}(Y_t) = \sigma^2$ does not depend on t
3. $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ depends only on j but not on t, for $j = 1, 2, ...$